

Requirement Analysis

Solution Requirement (Functional & Non-Functional)

Date	28-09-2026
Team ID	SWTID-2026-8620
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Functional Requirement:

This system is used to import data from Excel or CSV files. First it uploads and checks the file and loads it into a staging table. Then it uses a Transform Map to match the Excel columns with the target table columns like CMDB or Incident. It updates old records and adds new records. If any row fails, it will save the error and show it. Admin can see all import history and download error reports.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import from Spreadsheet	• Upload Excel/CSV • Validate file format • Load into Import Set Table (Staging)
FR-2	Transform Map Configuration	• Create Transform Map • Map source fields to target fields (CMDB/Incident) • Set Coalesce logic
FR-3	Data Transformation	• Transform data to target table • Auto-update existing records • Insert new records
FR-4	Error Handling & Logs	• Log failed records • Display error messages • Allow re-import of failed rows
FR-5	Admin Dashboard	• View import history • View success/failure count • Download error log

Non -Functional Requirement:

The system should be easy to use, fast, and secure. It should upload 10,000 rows in 2 minutes. Only admin can import data. Data should be 99% correct with no loss. It should work for many tables and support both Excel and CSV. It should be online always and save logs for 30 days.

NFR No.	Non-Functional Requirement (NFR)	Description
NFR-1	Usability	Simple UI for uploading spreadsheet, clear error messages for user.
NFR-2	Performance	Should handle 10,000+ rows in less than 2 minutes, fast transformation.
NFR-3	Security	Only admin can import, data validation to prevent injection.
NFR-4	Reliability	99% data accuracy after transform, no data loss during import.
NFR-5	Scalability	Support multiple target tables (CMDB, Incident, User). Support Excel and CSV formats.
NFR-6	Availability	System available 99.9%, import logs retained for 30 days.